

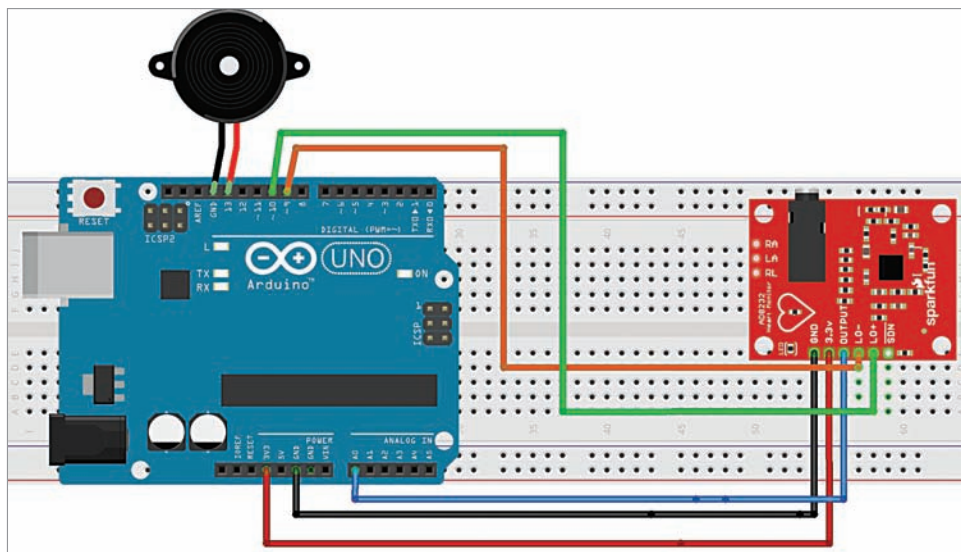


Fig. 4: Wiring of the system on a breadboard

```

Arduino Code:
int ecg = 0;
int buzzer = 11;
int Signal;
int Threshold = 590;
void setup()
{
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
  pinMode(9, INPUT);
  pinMode(10, INPUT);
}
void loop()
{
  if((digitalRead(9) == 1) || (digitalRead(10) == 1))
  {
    Serial.println("!");
  }
  Signal = analogRead(ecg);
  Serial.println(Signal);
  if(Signal > Threshold)
  {
    tone(buzzer, HIGH);
  }
  Else
  {
    noTone(buzzer);
  }
  delay(12);
}

```

Fig. 5: A snapshot of the source code for ECG sensor

shows its block diagram.

As shown in Fig. 2, a low-cost real-time ECG machine is proposed here using Arduino and a single lead. It also gives an alarm when the signal exceeds a pre-decided threshold value. The setup consists of three body electrodes (LA goes to the left arm, RA to the right arm, and RL to the right leg).

Circuit diagram of the real-time handheld ECG measurement system is shown in Fig. 3 and its wiring on a breadboard is shown in Fig. 4. The system is built around an AD8232 sensor (MOD1), Arduino Uno (MOD 2), 3M ECG adhesive type electrode, and a few other components.

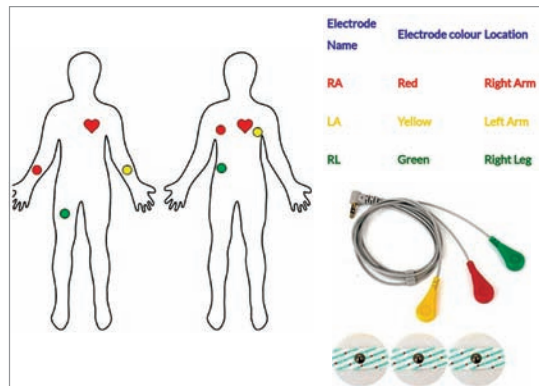


Fig. 6: Lead positions on a human body and colour code of the electrode connectors

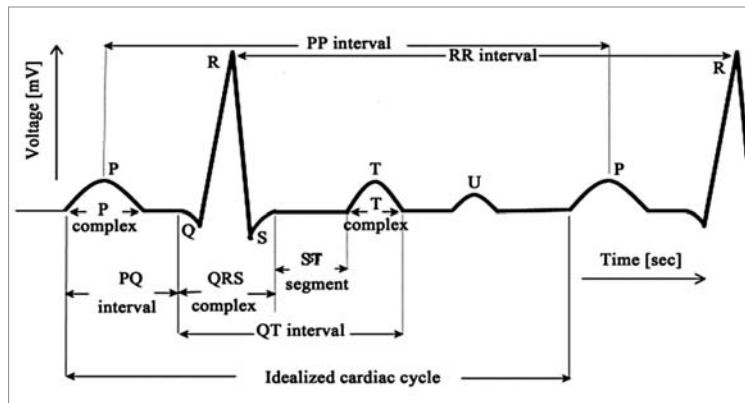


Fig. 7: A typical healthy ECG to understand the PQRSTU cardiac cycle

Arduino code ECG_code.ino is created and uploaded to the board by selecting the right port and board type in Arduino IDE. After that the AD8232 is connected according to circuit diagram in Fig. 4.

Construction and testing

To obtain a standard ECG, electrodes are attached to the three vital points on the person. The most common configuration, called the single-lead configuration, is used here, where both wrists and the right leg are connected through electrical leads for monitoring the functioning of the heart. Fig. 6 shows the preferred locations on a human body and colour code of the electrode connectors.

Bio potential electrodes are crucial in monitoring ECG signals. The right type of electrodes last for a long time and assist in recording high-quality signals. These low-cost, easily-available electrodes are made of Ag/AgCl and are surrounded by a conductive gel whose resistivity is about $100\Omega\cdot\text{m}$. The gel is surrounded by an adhesive material that is used to fix the electrodes.

The ECG graph contains repetitive cycles, and each cycle contains